

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall get GPS coordinates from active city buses every 5 seconds and update their current locations.	High	The GPS contains updated data of all the current locations.	Required for preparation .
FR-002	Functional	The system shall calculate and display estimated arrival times (ETA) for buses at upcoming route stops.	High	ETA is calculated based on current traffic information and road information and displayed to customers.	Commuters need accurate information about the ETA of buses.
FR-003	Functional	The system shall compute passenger crowding levels using data received from ticketing sensors.	High	The system calculates and displays the current crowding level for an active bus.	Crowd information helps commuters choose suitable buses and helps controllers monitor capacity.
FR-004	Functional	The system shall allow a commuter to view live bus locations, ETA information, and crowding levels for	High	A commuter can select a bus or route and view its current location, ETA, and crowding level.	Provides commuters with the live transport information they need.

		buses and route stops.			
FR-005	Functional	The system shall allow the Fleet Controller to monitor active buses and their operational status through the tracking platform.	High	The Fleet Controller can view active buses and their latest tracking information.	Fleet monitoring supports effective management of the municipal bus service.
NFR-001	Non-Functional	The tracking engine shall handle telemetry streams from up to 1,000 active transit vehicles simultaneously.	High	Benchmarking confirms that the system continues processing telemetry within the required performance limits for 1,000 active vehicles.	The platform must remain usable during peak periods with many active buses.
NFR-002	Non-Functional	The system shall protect bus telemetry and passenger-related sensor data from unauthorized access or modification.	High	Access-control and security tests confirm that unauthorized users cannot access or modify protected data.	Protecting operational and passenger-related data is essential for a secure transit platform.